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Data Warehousing and Business Intelligence

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6.6 Data Visualization

What is Data Visualization?



Overview of visualization, why it's essential for data comprehension.

- **Transforming data into visual insights - Turning raw data into visual forms.**
- **Enhances data accessibility, comprehension, and usability.**
- **Essential for Business Intelligence (BI) applications and Decision Support Systems (DSS).**

Why It Matters:

- **Simplifies complex data.**
- **Drives data-driven decisions.**



Choosing the Right Visualization

Audience and Purpose

- Understand who will view the data and what decisions they need to make.
- Ensure the visualization speaks clearly to their goals.

Data Relationships

- Comparison: Bar charts, column charts.
- Trends: Line graphs, area charts.
- Composition: Pie charts, stacked bar charts.
- Distribution: Histograms, scatter plots.

Data Complexity

- Choose visuals that simplify complex data.
- Avoid over-complicated visuals that may overwhelm.

Key Questions to Ask:

- What story does the data need to tell?
- Which visual form makes insights easiest to interpret?
- Can this type handle the data volume effectively?

Tools and Technologies (Overview)

- Tableau: *Advanced dashboards, visuals.*
- Power BI: *Integrates with Microsoft tools.*
- QlikView: *Interactive visual analysis.*
- D3.js: *Customizable, web-based visuals.*
- Google Data Studio: *Easy integration with Google data sources.*
- Excel: *Versatile, widely used.*

Data Sources for Visualization



- **Relational Databases**
- **Data Warehouses**
- **Cloud Data Sources**
- **APIs**
- **Spreadsheets**

Connecting Data: Data extraction and integration into visualization tools.

Design Principles – Visual Clarity and Accessibility



- **Color Theory:** Contrast, consistency.
- **Typography:** Readable fonts, hierarchy.
- **Layout & Composition:** Balanced, organized flow.
- **Accessibility:** Visual accommodations (e.g., color-blind options).

Design Principles – User Experience (UX)



- **Intuitive navigation.**
- **Consistent visual cues.**
- **Minimalist approach – avoid clutter.**
- **Consider audience needs.**

Activity – Critique a Visualization

- Choose a visual (from real-world data).
- Analyze using design principles.
- Discuss: What works? What could improve?

Analytics and Insights (Overview)

Types of Analytics

- **Descriptive: What happened?**
- **Predictive: What might happen?**
- **Prescriptive: What should be done?**
- **KPIs and Metrics: Tracking progress.**
- **Trend Analysis: Identifying patterns.**

Business Applications of Data Visualization



Applications

- Sales and Marketing Analytics
- Financial Reporting
- Operational Efficiency
- Customer Insights
- Supply Chain Management

Interactivity in Data Visualization



Purpose: Engages users, enables deeper analysis.

- **Drill-down Capabilities**
- **Filtering and Slicing**
- **Tooltips and Annotations**
- **Real-time Data Updates**

Discussion

Think of a scenario in your own industry or a familiar context.

Identify a Fact and two Dimensions you would analyze (e.g., total transactions for a banking service across branches and time periods).

Challenges and Considerations in Visualization



Key Challenges

- **Data Quality and Accuracy**
- **Overloading Information**
- **Security and Privacy**
- **Scalability**
- **Integration with Existing Systems**

Future Trends in Data Visualization



Emerging Trends

- **AI and Machine Learning**
- **AR and VR Applications**
- **Real-time Visualization**
- **Predictive and Prescriptive Visualizations**
- **Mobile Data Visualization**

Summary and Best Practices



- Match visual type to data and audience.
- Prioritize clarity, simplicity, accessibility.
- Use interactivity to engage and deepen insights.
- Stay updated on trends and tool advancements.

Questions and Discussion

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